1.6 Electrochemistry

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Hard

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

This question is about the tetrathionate ion.

The tetrathionate ion is shown in Fig 1.1.

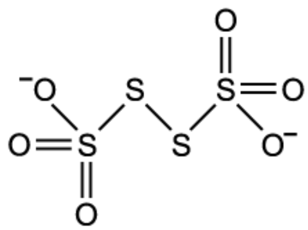


Fig 1.1

Calculate the oxidation number of sulfur in the tetrathionate ion. Explain your answer.

[2 marks]

Question 1b

Sodium tetrathionate can be formed by reacting sodium thiosulfate, $Na_2S_2O_3$, with iodine.

i)

Suggest an equation for the reaction between sodium thiosulfate and iodine.

[2]

ii)

Identify the oxidising agent in this reaction.

[1]

iii)

Describe the expected observation to show that this reaction had gone to completion.

[1]

[4 marks]

Question 1c

Sodium tetrathionate can also be made by reacting sodium bisulfite, $NaHSO_3$, with disulfur dichloride, S_2Cl_2.

$$2NaHSO_3 + S_2Cl_2 \rightarrow Na_2S_4O_6 + 2HCl$$

Explain if this reaction can be described as a disproportionation reaction. Include reference to oxidation numbers of sulfur in your answer.

[3 marks]

Question 2a

Photochromic glass contains an evenly distributed mixture of copper(I) chloride and silver(I) chloride. When sunlight passes through the glass the silver(I) chloride is separated into its ions. The chloride ions are then converted into chlorine atoms and the silver(I) ions into silver atoms. The silver atoms cluster together causing the lenses of photochromic glasses to darken.

Write equations for the processes involved in the darkening of photochromic glasses and explain if each reaction is reduction or oxidation.

[3 marks]

Question 2b

Prescription photochromic sunglasses react to ultraviolet light and are adjusted to the needs of the wearer.

Explain why someone wearing photochromic sunglasses might not receive the full benefit of the glasses when driving a car on a cold, bright day.

[2 marks]

Question 2c

The darkening process caused by the formation of silver atoms is reversible.

Write two equations to show how copper(I) chloride can react with the products from part (a) to remove the silver atoms and cause the lens to lighten.

[2 marks]

Question 2d

Copper(I) chloride, similar to that used in photochromic glass, can be made in the laboratory by the following method:

- Warm 0.5 g copper(II) oxide with 5 cm^3 concentrated hydrochloric acid for 1 minute.
- Add 1.0 g of copper turnings.
- Gently boil for 5 minutes.
- Filter the solution into 250 cm^3 of deionised water.
- Allow the copper(I) chloride precipitate to settle.
- Decant the liquid.
- Allow the copper(I) chloride to dry.

i)

Write an equation, including state symbols, for the reaction described.

[1]

ii)

Explain, using oxidation numbers, if this is a redox, disproportionation reaction.

[2]

[3 marks]

Question 3a

The reaction of sodium iodide with concentrated sulfuric acid forms a variety of products as shown in Table 3.1.

Table 3.1

Chemical	Formed by
$NaHSO_4$	
SO_2	
Na_2SO_4	
I_2	
HI	
S	
H_2O	
H_2S	

Complete Table 3.1 by identifying which products are formed by oxidation or reduction.

[3 marks]

Question 3b

Sulfur dioxide reacts with a solution of copper(II) chloride.

$$SO_2\ (g) + 2H_2O\ (l) + 2CuCl_2\ (aq) \rightarrow H_2SO_4\ (aq) + 2HCl\ (aq) + 2CuCl\ (s)$$

Identify the reducing agent and oxidising agent in this reaction.

[2 marks]

Question 3c

Sodium iodate is one of the main sources of iodine in the world. To extract the iodine, sodium iodate is reacted with sodium hydrogen sulfite, $NaHSO_3$, according to the following ionic equation.

$$2IO_3\ (aq) + 5HSO_3^-\ (aq) \rightarrow 3HSO_4^-\ (aq) + 2SO_4^{2-}\ (aq) + I_2\ (aq) + H_2O\ (l)$$

i)

Explain why sodium does not appear in the equation.

[1]

ii)

Explain the role of the hydrogen sulfite ions in the reaction.

[2]

[3 marks]

Model Answers: Hard

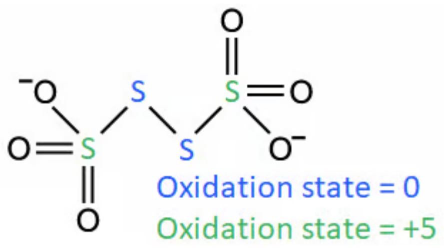
1a

a) The oxidation number of sulfur in the tetrathionate ion is:

- +2.5; [1 mark]
- (Because) the oxidation number is an average of the different oxidation numbers of sulfur
  OR
  (Because) the sulfur has two different oxidation numbers; [1 mark]

[Total: 2 marks]

- To calculate the oxidation number of S in $S_4O_6^{2-}$:
  - Each oxygen atom has an oxidation number of –2 making –12 in total
  - The ion has an overall oxidation number of –2
  - This means that the sulfur atoms must balance out –10
  - The 4 sulfur atoms must be +10, therefore each sulfur atom has an oxidation number of +2.5



- Don't be put off by an oxidation number value that isn't a whole number
  - When the oxidation number isn't a whole number, it is always good to double-check your answer
  - Although it is rare in exams, this can happen when you are actually getting the average of different oxidation numbers

1b

b)

i) The balanced symbol equation for the reaction of sodium thiosulfate with iodine to form sodium tetrathionate is:

$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

- Correct formula for sodium tetrathionate, $Na_2S_4O_6$
  AND
  Correct identification of the other product, NaI; [1 mark]
- Equation correctly balanced; [1 mark]

ii) The oxidising agent in this reaction is:

- I_2 / iodine; [1 mark]

iii) The expected observation to show that this reaction had gone to completion is:

- The discolouration of I_2 / iodine
  OR
  The reaction mixture / iodine turning from orange / red-brown to colourless; [1 mark]

[Total: 4 marks]

- Part (i) can be off-putting because it doesn't tell you all of the chemicals involved in the reaction
  - You have to deduce the formula of sodium tetrathionate using the information from part (a)
  - One product (NaI) is missing from the information given
- Part (ii) does not need you to look at oxidation numbers
  - The sodium thiosulfate has gained oxygen, which is one of the definitions of oxidation
- Part (iii) requires you to know the colour of iodine as orange / brown / red-brown
  - **Careful:** Do not write just red on its own as you will probably lose the mark

1c

c) Explanation of whether this reaction can be described as disproportionation:

The oxidation numbers of sulfur are:

- In $NaHSO_3$ = +4

AND
In S_2Cl_2 = +1; [1 mark]

The reaction between sodium bisulfite and disulfur dichloride:

- Is not a disproportionation reaction / is not disproportionation
  AND
  Because disproportionation involves one chemical / reactant / single substance undergoing reduction and oxidation at the same time / simultaneously; [1 mark]

Any **one** of the following:

- (Because) the oxidation number of sulfur in $NaHSO_3$ decreases which is reduction **AND** the oxidation number of sulfur in S_2Cl_2 increases which is oxidation
  OR
  (Because) the (sulfur in) $NaHSO_3$ is reduced but (sulfur in) S_2Cl_2 is oxidised; [1 mark]

[Total: 3 marks]

- The oxidation number of S in $NaHSO_3$
  - The oxidation of oxygen = -2
  - The oxidation number of sodium = +1 and hydrogen = +1
  - $NaHSO_3$ is overall neutral so the oxidation number of S must be +4
- The oxidation number of S in S_2Cl_2
  - The oxidation of chlorine = -1
  - S_2Cl_2 is overall neutral so the oxidation number of S must be +1
- The oxidation number of $Na_2S_4O_6$ is not required for the mark but if $Na_2S_4O_6 \neq +2.5$ then the first mark is lost
- **Remember:** A disproportionation reaction is one in which a single substance is simultaneously oxidised and reduced

2a

a) The equations for the processes involved in the darkening of photochromic glasses are:

- $AgCl \rightarrow Ag^+ + Cl^-$
  AND
  This is not reduction or oxidation as there is no change in oxidation states; [1 mark]
- $Cl^- \rightarrow Cl + e^-$

AND
This is an oxidation reaction as the chloride ions are losing electrons; [1 mark]

- $Ag^+ + e^- \rightarrow Ag$
  AND
  This is a reduction reaction as the silver ions are gaining electrons; [1 mark]

[Total: 3 marks]

- **Careful:** It is easy to lose marks on this question
- The second half of the question asks you to explain if each reaction is reduction or oxidation
  - There are 3 reactions stated in the question but only 2 of them are reduction / oxidation
  - You should include all of the reactions as the question asks for this
- The question states "into chlorine atoms" so your equation should form Cl not Cl2
- **Remember:** When talking about electrons, it's all about OIL RIG - oxidation is loss, reduction is gain

2b

b) Someone wearing photochromic sunglasses might not receive the full benefit of the glasses when driving a car on a cold, bright day because:

- (On a cold day) the glasses / lenses will be slow to darken / take time to darken; [1 mark]
- (In a car) the car / glass / windscreen will absorb some of the UV light so they won't darken as much; [1 mark]

[Total: 2 marks]

- Sometimes when you are faced with a question in an unfamiliar situation, you have to rely on your exam technique
- The unusual points of this question are "driving a car on a cold, bright day"
  - "Bright" doesn't help because the glasses are designed to react to light / UV
  - "Cold" would slow the rate of reaction down, which wouldn't be beneficial
  - "Driving in a car" - you have to consider how the car can affect the light / UV and the only way it can do this is by the car / glass / windscreen absorbing some of the light / UV

2c

c) Copper(I) chloride is involved in the lightening process by:

- $CuCl + Cl \rightarrow CuCl_2$
  OR
  $CuCl + Cl \rightarrow Cu^{2+} + 2Cl^{-}$; [1 mark]
- $CuCl_2 + Ag \rightarrow CuCl + AgCl$
  OR
  $Cu^{2+} + Ag \rightarrow Cu^{+} + Ag^{+}$; [1 mark]

[Total: 2 marks]

- A quick answer to this question would be to suggest that the silver displaces the copper in copper(I) chloride but this cannot happen because:
  - Copper is above silver in the reactivity series
  - If this did happen then you would be left with copper making the lenses a red-orange colour
- Think about the chemical species you have available:
  - Copper(I) chloride, CuCl
  - Silver atoms, Ag
  - Chlorine atoms, Cl
- You need to have copper(I) chloride, silver ions and chloride ions at the end, but silver is unreactive:
  - Reacting the copper(I) chloride with the chlorine atoms produces Cu^{2+} and chloride ions
  - A redox reaction between the Cu^{2+} ions and Ag atoms produces Cu^{+} and Ag^{+}

orange colour

- Think about the chemical species you have available:
  - Copper(I) chloride, CuCl
  - Silver atoms, Ag
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  - A redox reaction between the Cu^{2+} ions and Ag atoms produces Cu^{+} and Ag^{+}

2d

d)

i) The balanced symbol equation, including state symbols, for the formation of copper(I) chloride is:

- CuO (s) + Cu (s) + 2HCl (aq) → 2CuCl (s) + H_2O (l); [1 mark]

ii) Using oxidation numbers, the reaction is:

- A redox reaction / redox
  AND
  (Because) the oxidation state of copper changes from +2 in CuO and 0 in Cu to become +1 in CuCl; [1 mark]
- Not a disproportionation reaction / not disproportionation
  AND
  (Because) there isn't an element in just one of the reactants that undergoes reduction and oxidation at the same time / (Because) the copper that undergoes reduction and oxidation is in two different chemicals; [1 mark]

[Total: 3 marks]

- **Careful:** The question specifically states to include state symbols, so without these you cannot get the mark
  - **Remember:** HCl is aqueous not a liquid
- By definition, a disproportionation reaction is a redox reaction
  - The word redox is there as a deliberate distractor

- It is included to make you focus on reduction and oxidation
- It is included to make you ignore that disproportionation is reduction and oxidation of an element in one chemical at the same time

3a

a) The products in the table are formed by:

Chemical	Formed by	Chemical	Formed by
$NaHSO_4$	Neither	HI	Neither
SO_2	Reduction	S	Reduction
Na_2SO_4	Neither	H_2O	Neither
I_2	Oxidation	H_2S	Reduction

- All reduction reactions correctly identified; [1 mark]
- All oxidation reactions correctly identified; [1 mark]
- All reactions correctly identified; [1 mark]

[Total: 3 marks]

- Take your time with this question to ensure you get all of the oxidation numbers correct
  - Hydrogen oxidation number = +1 throughout
  - Sodium oxidation number = +1 throughout
  - Oxygen oxidation number = -2 throughout
  - Deal with the SO_4^{2-} as the ion with an oxidation number of -2 where possible
- Having established these oxidation numbers, you are left with sulfur and iodine as the elements to review in more detail
  - Sulfur in H_2SO_4 = +6
  - Sulfur in SO_2 = +4
  - Elemental sulfur = 0
  - Iodine in NaI and HI = -1
  - Elemental iodine = 0
- **Remember:** OIL RIG
  - Oxidation is the loss of electrons(and thus an increase in oxidation state)

    - Reduction is the gain of electrons (and thus a decrease in oxidation state)

3b

b) The reducing and oxidising agent in this reaction are:

- Reducing agent = Sulfur dioxide / SO_2; [1 mark]
- Oxidising agent = Cu^{2+} ions; [1 mark]

[Total: 2 marks]

- Using oxidation numbers, the only elements that have a change in oxidation number are sulfur and copper
- **Remember:** A reducing agent causes reduction and is oxidised itself / has an increase in oxidation number:
    - Sulfur changes from +4 in SO_2 to +6 in H_2SO_4
    - This means that the SO_2 is the reducing agent because it contains the sulfur that is being oxidised
- **Remember:** An oxidising agent causes oxidation and is reduced itself / has a decrease in oxidation state:
    - Copper changes from +2 in $CuCl_2$ to +1 in CuCl
    - The actual oxidising agent is the Cu^{2+} ions as they are being reduced / gaining electrons
        - The $CuCl_2$ is not the oxidising agent

3c

c)

i) Sodium does not appear in the equation because:

- It is a spectator ion
  OR
  Sodium does not take part in the reaction; [1 mark]

ii) The role of the hydrogen sulfite ions is:

- Reducing agent / reductant; [1 mark]
- (Because) the sulfur changes from an oxidation number of +4 in the HSO_3^- ions to an

oxidation number of +6 in the HSO_4^- and SO_4^{2-} ions

OR

(Because) the sulfur is oxidised from +4 in the HSO_3^- ions to +6 in the HSO_4^- and SO_4^{2-} ions; [1 mark]

[Total: 3 marks]

- **Remember:** Spectator ions don't actually participate in the chemistry of a reaction:
  - This means that you don't necessarily have to include them in a chemical equation
- Make sure to check the question:
  - If they ask for an ionic equation, you can omit the spectator ions
  - If they ask for a full / overall equation, then you must include the spectator ions.
- Ionic equations often look more complicated than they actually are
- The question narrows the ionic equation down to the hydrogen sulfite ions:
  - You know that hydrogen has an oxidation state of +1 (except hydrides)
  - You know that oxygen has an oxidation state of -2 (except peroxides)
  - You can deduce that the sulfur has an oxidation state of +4
- The only important parts of the ionic equation are
  - $5HSO_3^-\ (aq) \rightarrow 3HSO_4^-\ (aq) + 2SO_4^{2-}\ (aq)$
- As sulfur is the only element with a variable oxidation state then the whole question depends on the oxidation state of sulfur in the products
  - **Careful:** Hydrogen sulfite ions are HSO_3^-
  - It is easy to automatically go to HSO_4^- because this is an ion that is more familiar to you (from sulfuric acid, H_2SO_4)